

Patient Profile

Saubhik Bhaumik, 27-year-old male

Summary of Results

Liver function tests demonstrate values entirely within reference ranges, indicating preserved hepatic function. **ML Result: Normal** – the predictive model assesses the profile as within normal limits.

Detailed Test Explanations

Serum Bilirubin (Total): Measures overall bilirubin, a byproduct of red blood cell breakdown processed by the liver; elevated levels suggest hemolysis, liver dysfunction, or biliary obstruction. Patient result: 0.9 mg/dL (reference: 0.2-1.2 mg/dL) – normal.

Serum Bilirubin (Direct): Reflects conjugated bilirubin excreted in bile; elevations indicate hepatocellular or obstructive liver disease. Patient result: 0.2 mg/dL (reference: 0-0.3 mg/dL) – normal.

Serum Bilirubin (Indirect): Represents unconjugated bilirubin; increases may signal hemolysis or impaired hepatic conjugation. Patient result: 0.70 mg/dL (reference: 0.2-1.0 mg/dL) – normal.

SGPT (ALT): Alanine aminotransferase, a liver enzyme highly specific for hepatocellular injury; elevations denote acute liver damage. Patient result: 36 U/L (reference: 13-40 U/L) – normal.

SGOT (AST): Aspartate aminotransferase, present in liver, heart, and muscle; rises indicate tissue damage, often hepatic. Patient result: 32 U/L (reference: 0-37 U/L) – normal.

Serum Alkaline Phosphatase: Enzyme from liver, bone, and intestine; elevations suggest cholestasis, bone disease, or infiltration. Patient result: Not available.

Serum Protein: Total protein reflects synthetic liver function and nutritional status; abnormalities indicate chronic liver disease or malnutrition. Patient result: 7.2 g/dL (reference: 6.4-8.3 g/dL) – normal.

Serum Albumin: Primary protein synthesized by liver; hypoalbuminemia signals chronic liver disease or inflammation. Patient result: 4.7 g/dL (reference: 3.5-5.2 g/dL) – normal.

Globulin: Immunoglobulin fraction; derangements may indicate infection, autoimmune disease, or liver dysfunction. Patient result: 2.50 g/dL (reference: 1.8-3.6 g/dL) – normal.

A/G Ratio: Albumin-to-globulin ratio assesses protein balance; imbalances suggest chronic inflammation or liver synthetic defects. Patient result: 1.88 (reference: 1.1-2.4) – normal.

Overall Interpretation

Comprehensive liver function panel shows all parameters within physiological norms, consistent with intact hepatocellular integrity, bilirubin metabolism, and protein synthesis. The absence of alkaline phosphatase precludes full profiling but does not alter the normal pattern. **ML Result: Normal** – predictive model confirms profile within normal limits, with no evidence of hepatic pathology.

Recommendations

Maintain balanced diet rich in fruits, vegetables, and lean proteins; engage in regular aerobic exercise (150 minutes weekly); limit alcohol intake; avoid hepatotoxic agents. Routine annual hepatic screening advised; consult physician for personalized follow-up or symptoms such as fatigue or jaundice.

Disclaimer

This report provides interpretive guidance based on provided data and is not a substitute for professional medical evaluation, diagnosis, or treatment.